

# The Digital Age of HP2020

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# This Morning's Agenda

Overview of Healthy People

.NET to Drupal Migration

User Experience Improvement

Key Features and Drupal Taxonomies

Next Steps



# What Is Healthy People?

- Provides **science-based, 10-year national objectives** for improving the health of the Nation
- Communicates a vision for improving health and achieving health equity on the national level
- Identifies **measurable objectives** with **targets** to be achieved by the year 2020
- Requires tracking of **data-driven outcomes** to monitor progress and to motivate, guide, and focus action



# 42 Topic Areas

- Access to Health Services
- Adolescent Health
- Arthritis, Osteoporosis, and Chronic Back Conditions
- Blood Disorders and Blood Safety
- Cancer
- Chronic Kidney Disease
- Dementias, including Alzheimer's Disease
- Diabetes
- Disability and Health
- Early and Middle Childhood
- Educational and Community-Based Programs
- Environmental Health
- Family Planning
- Food Safety
- Genomics
- Global Health
- Health-Related Quality of Life and Well-Being
- Health Communication and Health Information Technology
- Healthcare-Associated Infections
- Heart Disease and Stroke
- Hearing and Other Sensory or Communication Disorders
- HIV
- Lesbian, Gay, Bisexual, Transgender Health
- Immunization and Infectious Diseases
- Injury and Violence Prevention
- Maternal, Infant, and Child Health
- Medical Product Safety
- Mental Health and Mental Disorders
- Nutrition and Weight Status
- Occupational Safety and Health
- Older Adults
- Oral Health
- Physical Activity
- Public Health Infrastructure
- Preparedness
- Respiratory Diseases
- Sexually Transmitted Diseases
- Sleep Health
- Social Determinants of Health
- Substance Abuse
- Tobacco Use
- Vision



# HealthyPeople.gov Online Tools and Resources

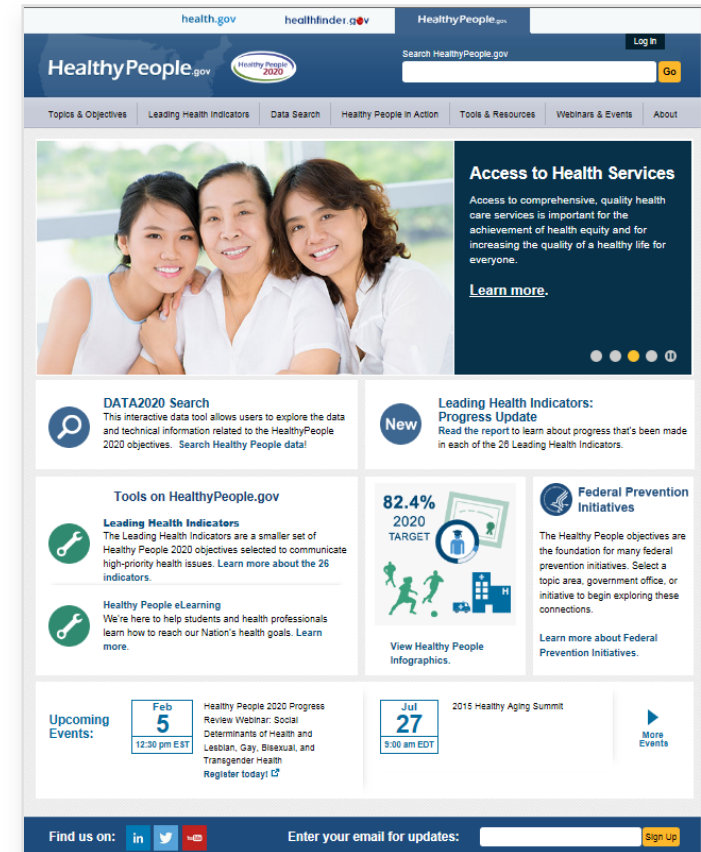
- Topic Area Information
- Objectives
- DATA2020
- National Snapshots
- Infographics
- Evidence-Based Resources (EBRs)
- Stories from the Field
- Social Determinants of Health (SDOH) Portal
- Content Syndication
- Federal Prevention Initiatives
- eLearning
- Webinars





# Content Management System: Migration to Drupal

- Fall 2013-Fall 2014: migration from custom CMS to Drupal CMS
- Opportunity to conduct usability testing and revisit existing content on the site
- ❖ Our perspective: we want to empower our users, so the refresh was about making sure that people who use HealthyPeople.gov are getting what they need out of it.





# Content Management System: Migration to Drupal

Over 2,000 pages migrated to Drupal 7!





# Keeping Users at the Center

Drupal migration provided opportunity to:

- Confirm and clarify user needs
- Refine feature requirements
- Update information architecture
- Improve overall user experience

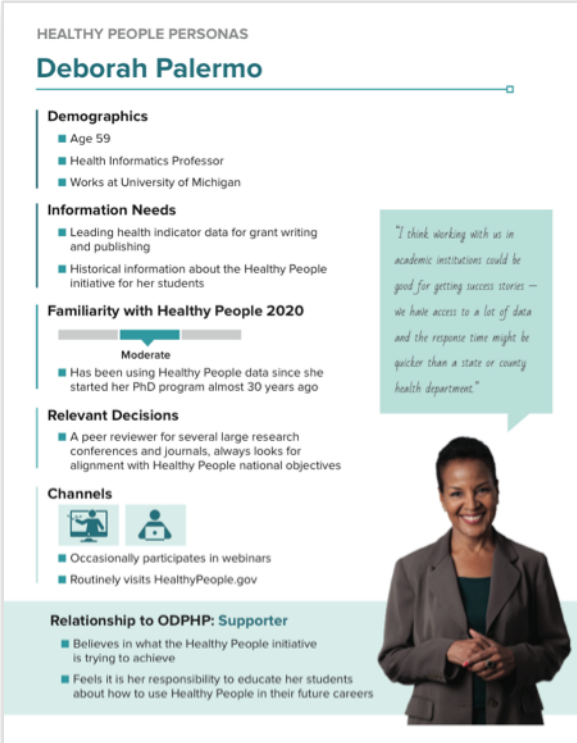


# User Research Approach

Gathered findings and recommendations using multiple methods:

- Focus groups and in-depth interviews
- Persona development
- Usability testing prior to Drupal migration
- Information architecture (tree testing)
- Prototype “mini-tests” during development of new tools and features
- Usability testing after Drupal migration to test updated features and new content

Required close integration of user research, content, and development teams



HEALTHY PEOPLE PERSONAS

## Deborah Palermo

**Demographics**

- Age 59
- Health Informatics Professor
- Works at University of Michigan

**Information Needs**

- Leading health indicator data for grant writing and publishing
- Historical information about the Healthy People Initiative for her students

**Familiarity with Healthy People 2020**

Moderate

- Has been using Healthy People data since she started her PhD program almost 30 years ago

**Relevant Decisions**

- A peer reviewer for several large research conferences and journals, always looks for alignment with Healthy People national objectives

**Channels**

- Occasionally participates in webinars
- Routinely visits HealthyPeople.gov

**Relationship to ODPHP: Supporter**

- Believes in what the Healthy People initiative is trying to achieve
- Feels it is her responsibility to educate her students about how to use Healthy People in their future careers

*"I think working with us in academic institutions could be good for getting success stories — we have access to a lot of data and the response time might be quicker than a state or county health department."*

# User Research Findings: DATA2020 Tool

## DATA2020 search tool

- Original in-depth interviews: Users are seeking public health data for specific HP2020 objectives
- Opportunity to redesign tool in Drupal using taxonomies and faceted search module — and address user needs



## Users struggled to use the tool on .NET

[Usability Testing Video: 0:46 - 1:56](#)

- Participants had difficulty narrowing down search results
- Multiple steps involved before users could see data in results

## Changes planned for Drupal

- Place filters on same page as results
- Provide real-time filtering and constant feedback
- Add advanced filtering to facilitate scanning

# Faceted Search Results

Reimagined search results pages using taxonomies and faceted search module



### Search HP2020 Objective Data by:

Topic Area Data Source

Select one or more Topic Areas to begin.

#### Objective Filters

67 Related Objectives [Reset](#)

- Topic Areas (0 Selected)
- Data Sources (0 Selected)
- State-Level Data Available (0 Selected)
- Sex (0 Selected)
- Age Group (1 Selected)
- Adolescents & Young Adults (10-24) (277)
- Adults (25-64) (204)
- Children (<1-9) (221)
- ☒ Older Adults (65 and older)
- Setting (0 Selected)
- Community (15)
- Criminal Justice System (0)
- Health Care Setting (33)
- School-based Setting (0)
- Worksite (0)
- Population Group (0 Selected)
- Intervention Agent (0 Selected)
- Prevention Areas (0 Selected)
- Type of Health Condition (0 Selected)
- LHI

#### Customize Display

☐ Show Confidence Interval (if available) ☐ Show Standard Error (if available) [Hide Populations](#)

☒ Display All Populations

☒ Total

☒ Sex

☒ Race/Ethnicity

☒ Age Group

☒ Educational attainment

☒ Family income

☒ Family type

☒ Country of birth

☒ Disability status

☒ Geographic location

☒ Health insurance status

☒ Sexual orientation

☒ Marital status

☐ Student/school characteristics

☒ Gender identity

☒ All other populations

[Refresh All Population Data](#)

67 Related Objectives [Download Search Results](#)

[Reset Search Criteria](#)

#### Access To Health Services

**AHS-5.4 Increase the proportion of adults aged 65 years and older who have a specific source of ongoing care**

Adults with a source of ongoing care (percent, 65+ years)

2020 Baseline (year): 96.3 (2008)    2020 Target: 100    Desired Direction: ↑ Increase desired

| POPULATIONS                      | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 |
|----------------------------------|------|------|------|------|------|------|
| TOTAL <a href="#">View Chart</a> | 96.3 | 96.6 | 96.5 | 96.3 | 96.3 | 96.3 |

[View Population Data](#)

National Data Source: National Health Interview Survey (NHIS), CDC/NCHS

Data: [Data Details](#) [Learn more about the methodology and measurement of this HP2020 objective](#)

Footnotes: [View All Footnotes](#)



# Drupal Taxonomies

Taxonomies applied to content types:

- DATA2020
- EBRs
- SDOH Portal

**SEARCH TERMS**  
**Age Group**  
- None -  
Children (&lt;1-9)  
Adolescents & Young Adults (10-24)  
Adults (25-64)  
**Sex**  
Female  
**Setting**  
- None -  
Community  
Criminal Justice System  
Health Care Setting  
**Population Group**  
- None -  
Caregivers  
Employers & Employees  
Parents, Families & Households  
**Intervention Agent**  
- None -  
Policymakers and Law Enforcement  
Public Health  
Academia, Research & Development  
**Prevention Areas**  
- None -  
Clinical Services  
Counseling  
Data Collection/Registries  
**Health Condition Type**  
- None -  
Chronic Conditions  
Communicable Diseases and Acute Infections



# Drupal Taxonomies

Taxonomies applied to content types:

- DATA2020
- EBRs
- SDOH Portal

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Target Population</b> <ul style="list-style-type: none"><li><input type="checkbox"/> Caregivers</li><li><input type="checkbox"/> Employers &amp; Employees</li><li><input checked="" type="checkbox"/> Families &amp; Households</li><li><input type="checkbox"/> Healthcare Providers</li><li><input type="checkbox"/> Homeless</li><li><input type="checkbox"/> LGBT</li><li><input type="checkbox"/> Low Income</li><li><input type="checkbox"/> Parents</li><li><input type="checkbox"/> Patients</li><li><input type="checkbox"/> Students</li></ul>                                                                         | <b>Resource Type</b> <ul style="list-style-type: none"><li><input checked="" type="checkbox"/> Non-Systematic Review</li><li><input type="checkbox"/> Systematic Review</li><li><input type="checkbox"/> Randomized Controlled Trial</li><li><input type="checkbox"/> Cohort Study</li><li><input type="checkbox"/> Cross-Sectional or Prevalence Study</li><li><input checked="" type="checkbox"/> Expert Opinion</li><li><input checked="" type="checkbox"/> Field-Based Summary or Case Study</li><li><input type="checkbox"/> Case-Control Study</li><li><input type="checkbox"/> Experimental Study</li><li><input checked="" type="checkbox"/> Other</li></ul> | <b>Intervention Type</b> <ul style="list-style-type: none"><li><input checked="" type="checkbox"/> Behavioral &amp; Social</li><li><input type="checkbox"/> Clinical Practice</li><li><input type="checkbox"/> Community Intervention</li><li><input type="checkbox"/> Environmental &amp; Policy</li><li><input type="checkbox"/> Informational &amp; Educational</li><li><input checked="" type="checkbox"/> Primary Prevention</li><li><input checked="" type="checkbox"/> Secondary Prevention</li><li><input type="checkbox"/> Social Marketing &amp; Media</li><li><input type="checkbox"/> Technology-based</li><li><input type="checkbox"/> Tertiary Prevention</li></ul> | <b>Desired Outcome</b> <ul style="list-style-type: none"><li><input checked="" type="checkbox"/> Effect Behavioral Change</li><li><input checked="" type="checkbox"/> Effect Health System Change</li><li><input type="checkbox"/> Eliminate Health Disparities</li><li><input checked="" type="checkbox"/> Improve Quality of Life</li><li><input type="checkbox"/> Reduce Injury</li><li><input checked="" type="checkbox"/> Reduce Morbidity</li><li><input type="checkbox"/> Reduce Mortality</li><li><input checked="" type="checkbox"/> Reduce Risk or Incidence of Disease or Condition</li></ul> | <b>EBR Setting</b> <ul style="list-style-type: none"><li><input type="checkbox"/> Clinical or Health Systems</li><li><input checked="" type="checkbox"/> Community</li><li><input type="checkbox"/> Criminal Justice System</li><li><input checked="" type="checkbox"/> Home</li><li><input type="checkbox"/> Medical Home</li><li><input type="checkbox"/> Rural</li><li><input type="checkbox"/> School</li><li><input type="checkbox"/> School-Based</li><li><input type="checkbox"/> Urban</li><li><input type="checkbox"/> Worksite</li></ul> |
| <b>EBR Intervention Agent</b> <ul style="list-style-type: none"><li><input type="checkbox"/> test</li><li><input checked="" type="checkbox"/> Public Health</li><li><input type="checkbox"/> Community- &amp; Faith-Based Organizations</li><li><input type="checkbox"/> Business &amp; Private Sector</li><li><input type="checkbox"/> Employers &amp; Employees</li><li><input checked="" type="checkbox"/> Government &amp; Policymakers</li><li><input type="checkbox"/> Law Enforcement</li><li><input checked="" type="checkbox"/> Healthcare</li><li><input type="checkbox"/> Academics, Research &amp; Development</li></ul> | <b>Health Condition Type</b> <ul style="list-style-type: none"><li><input type="checkbox"/> Chronic Conditions</li><li><input type="checkbox"/> Communicable Diseases &amp; Acute Infections</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Race and Ethnicity</b> <ul style="list-style-type: none"><li><input type="checkbox"/> American Indian or Alaska Native only</li><li><input type="checkbox"/> Asian only</li><li><input type="checkbox"/> Black or African American only</li><li><input type="checkbox"/> Hispanic or Latino</li><li><input type="checkbox"/> Native Hawaiian or Other Pacific Islander only</li><li><input type="checkbox"/> Non-Hispanic Black or African American</li><li><input type="checkbox"/> Non-Hispanic White</li><li><input type="checkbox"/> White only</li></ul>                                                                                                                  | <b>Age Ranges</b> <ul style="list-style-type: none"><li><input type="checkbox"/> &lt;1 year</li><li><input type="checkbox"/> 1-4 years</li><li><input type="checkbox"/> 5-9 years</li><li><input type="checkbox"/> 10-14 years</li><li><input type="checkbox"/> 15-19 years</li><li><input type="checkbox"/> 20-24 years</li><li><input type="checkbox"/> 25-44 years</li><li><input type="checkbox"/> 45-64 years</li><li><input type="checkbox"/> 65 years and older</li></ul>                                                                                                                         | <b>Sex</b> <ul style="list-style-type: none"><li><input type="checkbox"/> N/A</li><li><input type="checkbox"/> Sex</li><li><input type="checkbox"/> -Female</li><li><input type="checkbox"/> -Male</li><li><input type="checkbox"/> -Sex</li><li><input type="checkbox"/> Sex (of child)</li><li><input type="checkbox"/> -Female</li><li><input type="checkbox"/> -Male</li></ul>                                                                                                                                                                 |



# Drupal Taxonomies

Taxonomies applied to content types:

- DATA2020
- EBRs
- **SDOH Portal**

## Related Domains

- None -  
Economic Stability  
Education  
Health and Health Care

## Levels

- None -  
Local Resources  
State Resources  
National Resources



# SDOH Portal

User-friendly, visually pleasing front-end display for exposed faceted search results



The screenshot displays the SDOH Portal interface, featuring a top navigation bar with five colored tiles representing different domains: Economic Stability (orange), Education (blue), Health and Health Care (red), Neighborhood and Built Environment (green), and Social and Community Context (purple). Each tile contains a white icon: a dollar sign, a graduation cap, a medical cross, a house with a tree, and two people shaking hands, respectively. Below the tiles, a search filter bar shows "View content related to: All Domains" with a dropdown menu containing the same five domain icons. The main content area is titled "National, State, and Local Resources" and includes a sub-section "National Resources" with a list of links and descriptions. To the right of the list is a "Related Domains" column with icons corresponding to the domains. The list includes links to resources from the Robert Wood Johnson Foundation, ChangeLab Solutions, and the Smart Growth Network, each with a brief description and a PDF icon.

| Domain                             | Icon                     |
|------------------------------------|--------------------------|
| Economic Stability                 | Dollar sign              |
| Education                          | Graduation cap           |
| Health and Health Care             | Medical cross            |
| Neighborhood and Built Environment | House with tree          |
| Social and Community Context       | Two people shaking hands |

View content related to: All Domains

**National, State, and Local Resources**

Review these resources to see how others have worked to address and improve the social determinants of health across the Nation.

| National Resources                                                                                                                                                                                                                                                                    | Related Domains                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <a href="#">A New Way to Talk about the Social Determinants of Health (Robert Wood Johnson Foundation)</a><br>Evidence-based best practices in communicating about the social determinants of health                                                                                  | Economic Stability, Education, Health and Health Care, Neighborhood and Built Environment, Social and Community Context |
| <a href="#">A Roadmap for Healthier General Plans (ChangeLab Solutions) [PDF - 1.2 MB]</a><br>Toolkit on including health-promoting policies in local land use and development planning                                                                                               | Neighborhood and Built Environment                                                                                      |
| <a href="#">Division of Nutrition, Physical Activity and Obesity (DNPAO) State Program Highlights: Urban Design and Transportation Policies and Practices (CDC, Centers for Disease Control and Prevention) [PDF - 244 KB]</a><br>Examples of urban design policies across the nation | Neighborhood and Built Environment                                                                                      |
| <a href="#">Getting to Grocery: Tools for Attracting Healthy Food Retail to Underserved Neighborhoods (ChangeLab Solutions) [PDF - 1.0 MB]</a><br>Strategies to increase the availability of healthy grocery stores in your community                                                 | Neighborhood and Built Environment                                                                                      |
| <a href="#">Getting to Smart Growth: 100 Policies for Implementation (The Smart Growth Network) [PDF - 2.0 MB]</a><br>Framework to build healthy, vibrant, and diverse living environments through collaborative planning                                                             | Economic Stability, Neighborhood and Built Environment                                                                  |



# HHS Content Syndication Module

- Healthy People content syndicated into the HHS Content Syndication Storefront using the new HHS Content Syndication Drupal module
- New content is auto-published to the Syndication Storefront
- HealthyPeople content everywhere!





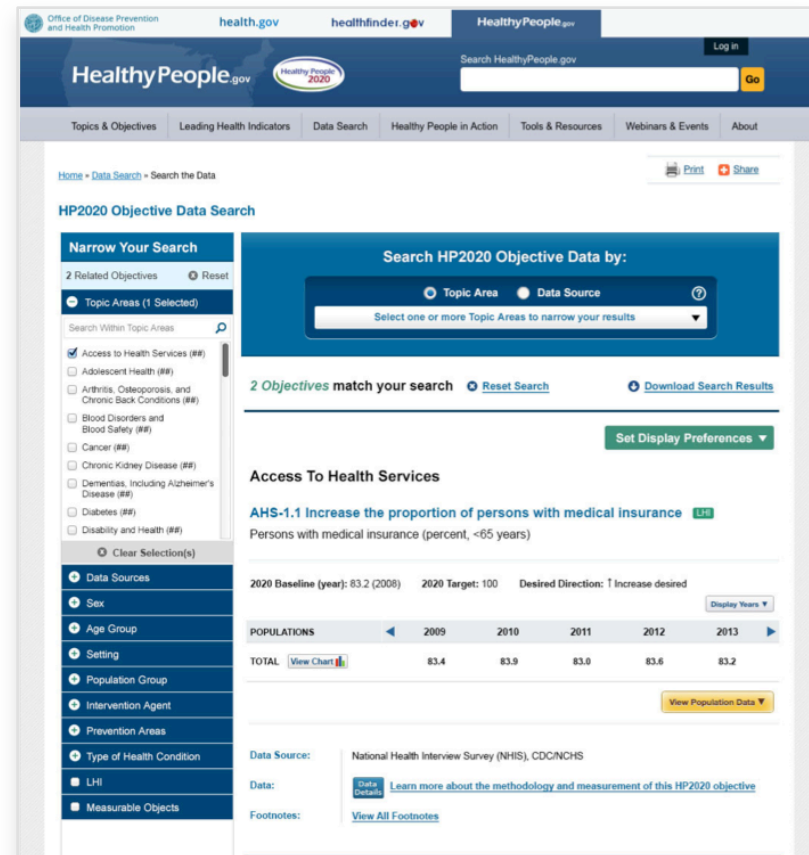
# Next Steps

## Drupal allows continual iterative improvements

- Expand CMS to expose DATA2020 entity fields to data managers
- Develop new DATA2020 search page layout to improve labels and scannability

## Ongoing testing as new content and features are added

- Segmented usability testing of general and DATA2020 features this summer





# Questions or comments?

## Office of Disease Prevention and Health Promotion

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## Stay Connected with HealthyPeople



Web

[HealthyPeople.gov](http://HealthyPeople.gov)



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Healthy People 2020